

Sean Berg

EDUCATION

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Carnegie Mellon University

M.S. in Biomedical Engineering (Machine Learning / AI)

Pittsburgh, PA

Aug. 2026 – Present

University of California, Davis

B.S. in Neurobiology, Physiology, and Behavior

Davis, CA

Sep. 2021 – Mar. 2024

AWARDS & RECOGNITION

Carnegie Mellon University

Biomedical Engineering Department Head's Fellowship; Academic Excellence Fellowship

Aug. 2026

WORK EXPERIENCE

Junior Specialist

The Silverman Translational Neuroscience Lab (PI: Jill Silverman)

Aug. 2024 – Aug. 2026

Sacramento, CA

- Engineered a novel behavioral assay integrating wireless EEG collection with operant conditioning touchscreen tasks; independently designed and fabricated custom acrylic platforms (SOLIDWORKS) and TTL-based synchronization system, saving >\$20K in vendor costs (n=12 mice).
- Built the longitudinal analysis pipeline for the above touchscreen-EEG dataset: trial-level ERP and time-frequency feature extraction (ERSP, ITPC), linear mixed-effects modeling, bootstrapped slope estimation, and cluster-based permutation testing across discrimination and reversal learning.
- Developed Python pipelines for analyzing continuous wireless EEG in awake behaving mice (ERP, multitaper spectrograms, 72-hour recordings); integrated within the core lab data analysis platform (involving approx. 100 mice/year).

Undergraduate Researcher

The Sensorimotor Integration Lab (PI: Wilsaan Joiner)

Aug. 2023 – Jul. 2024

Davis, CA

- Collected and analyzed sensorimotor trajectory data from force perturbation tasks (n=11 human subjects) to assess multisensory elements involved in short-term motor learning to inform forelimb prosthetic development; presented at the UC Davis 2024 Undergraduate Research Conference.

PUBLICATIONS & PRESENTATIONS

Manuscripts: *bioRxiv* preprint, under revision at *Brain*; 3 additional manuscripts in preparation (Shank3 ERP; translational ERP review; EEG-touchscreen methods)

Presentations: Poster, Society for Neuroscience Annual Meeting (2025); 3 institutional research showcases, UC Davis Health (2024–2026)

PROJECTS

Prototype Motor Imagery Brain-Computer Interface

Independent Project

Jul. 2025 – Feb. 2026

Davis, CA

- Built complete binary motor imagery BCI from hardware (AD620 instrumentation amplifier, TL084CN op-amp for analog signal filtering stages) to software (Python-based real-time signal processing and classification).
- Implemented synchronized PsychoPy task paradigm and logistic regression decoder with stratified k-fold cross-validation.

SKILLS & EXPERTISE

Programming: Python (NumPy, SciPy, scikit-learn, MNE-Python, pandas), MATLAB, Git, NEURON

Signal Processing & Statistics: ERP/ERSP/ITPC, Morlet and multitaper time-frequency decomposition, artifact rejection, spike detection, linear mixed-effects models, cluster-based permutation testing, cross-validated classification

Instrumentation: DSI wireless EEG telemetry, SOLIDWORKS, ABET II touchscreen, TTL synchronization, analog front-end design

In Vivo: Rodent stereotaxic surgery and EEG implantation (n=45, 100% survival), perfusions, histology, colony management, seizure induction, behavioral phenotyping battery (open field, novel object recognition, PPI, fear conditioning)